



Improve Payment Journey for E-commerce Application

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Executive Summary

Business Case

What is the importance of the project?

Customers are complaining that the payment process using the application takes time and hence, leads to customer frustration and loss of sales. The application ratings amongst the customers are dropping as well.

Root Cause Analysis

What are the critical findings/causes discovered?

- Payment portal loading time due to web-portal asset loading
- The payment processor response time is not known or communicated to the customer, leading to perceived delays and cancellations.
- Payments are completed, but a transaction response was not received from the payment process during the session.

Solutions Implemented

What are the solutions implemented to address root causes?

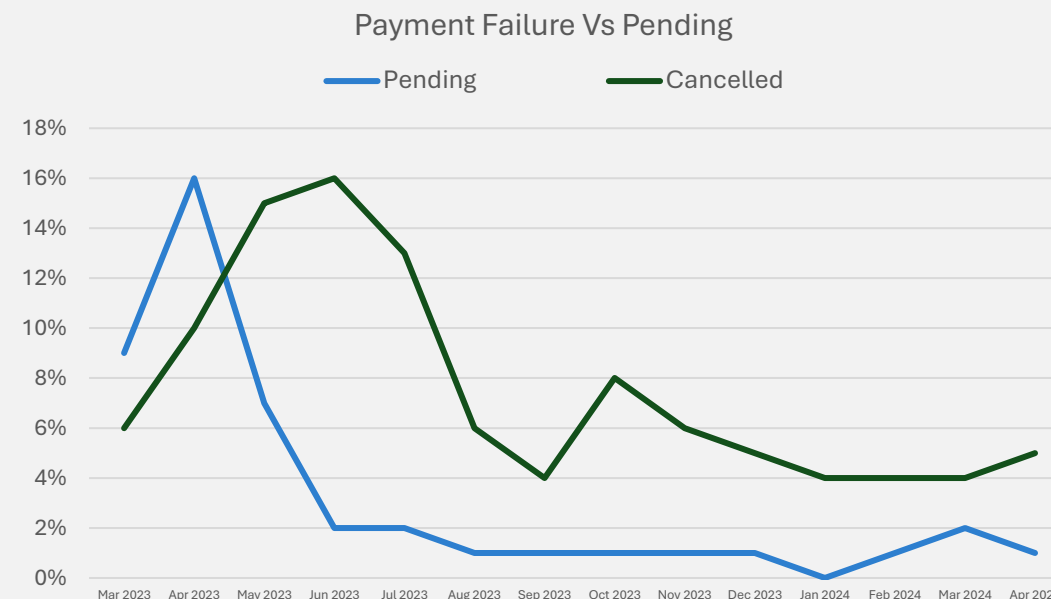
- Payment integration was changed from web-hosted checkout to native SDK integration provided by payment processor.
- Integrated with the payment process to share a callback for transactions when payment is approved.
- Added additional optimization of load times by removing reliance on server response and device initiates payments while informing server.
- Added additional visuals and communications during payment process to indicate status to the customers

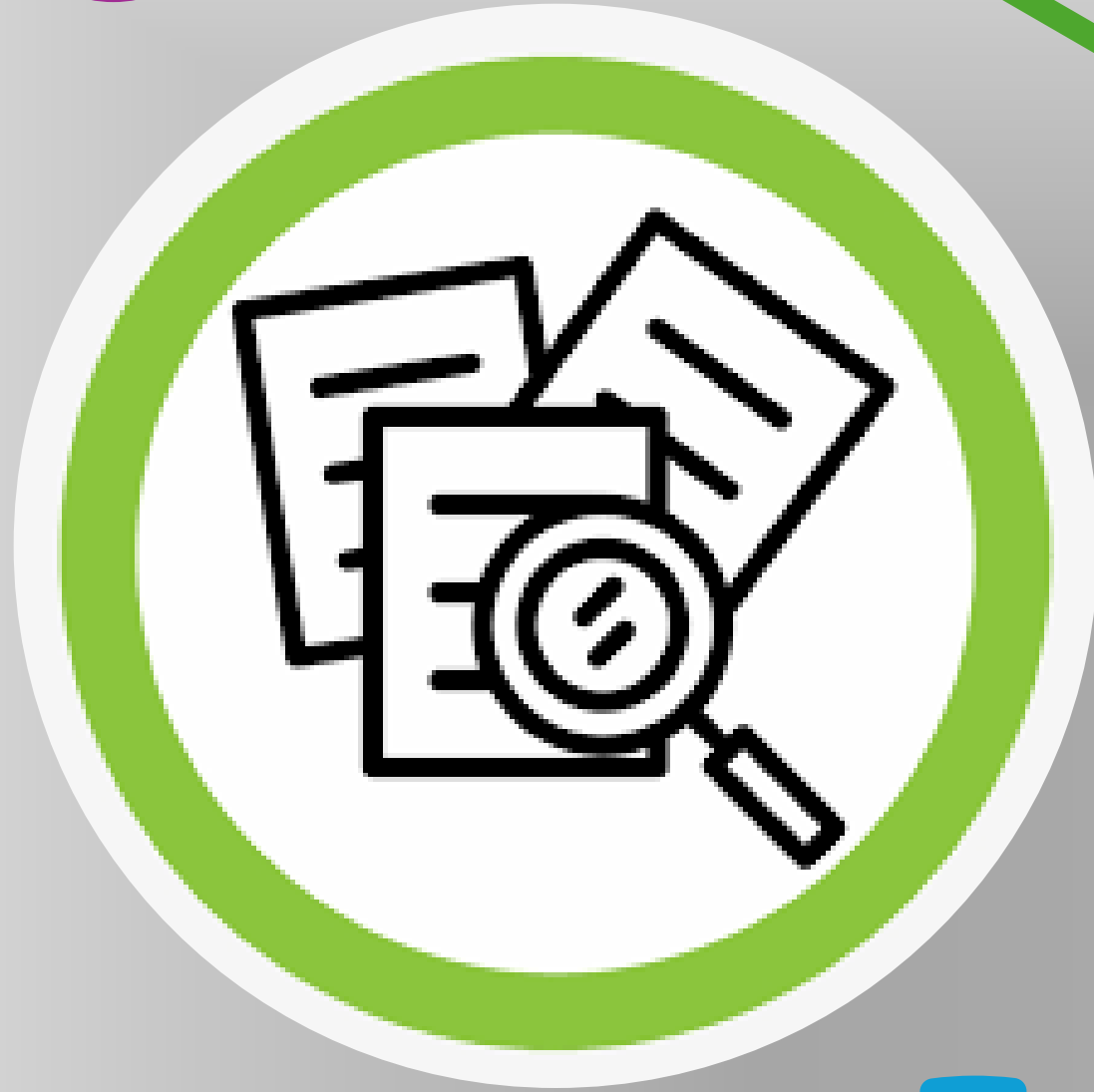
Project Results

What is the importance of the project?

- Decreased Payment pending (no response from payment processor) rate to avg 1.08%
- Decreased Customer Payment Cancellation rate to AVG 6% months.
- Avg Payment completion rate of 94%, above the industry benchmark of 85%.

Graphical Display of Improvement





Define Phase

Project Summary

Problem Statement

For the last 3 months, after the launch of the e-commerce product, customers have been facing difficulties during the payment stage of the Ordering journey, leading to loss of sales and negative customer reviews

Goal Statement

The platform should meet the industry standards for an e-commerce platform in terms of payment:

- “For e-commerce platforms, the average authorization rate typically ranges between 85% and 95%”- [Gr4vy](#)
- “The best optimized abandonment rate is approximately 20%”- - [Bolt](#)

Scope In/out

Process Start: Customer initiates payment by clicking checkout at the cart
Process End: Payment gateway returns the status of the payment.

In: Review of the payment journey with the payment gateway and processor.
Out: Review of the customer order journey in the application.

Business Case & benefits

Improving payment experience will lead to more transactions being completed resulting in more sales. This will increase the rating of the application by addressing the current difficulties faced by the customer

Timeline

Phase	Planned	Actual
Define	April 10 th	April 10 th
Measure	April 17 th	April 17 th
Analyze	April 24 th	April 27 th
Improve	May 1 st	May 2 nd
Control	May 22 nd	May 27 th

Team Members

Position	Person	Title	% of time
Team lead	Surya V	Greenbelt	40%
Sponsor	George P	Product Owner	10%
Team Member	Avinash T	Backend Developer	20%
Team member	Dipasri	Mobile Developer	20%
Team Member	Ceeliya L	Lead Tester	10%

Voice of the Customer Translation Matrix

Key Takeaway: Customers face friction with the adoption of the application due to payment.

Customer Comment What are they saying?	Context Why are they saying this?	Customer Requirement What do they want?
Customer transactions are not being accepted even after the amount is removed from their card	Response is not being received by the payment gateway due to the network drops or payment being cancelled before payment gateway responds with the actual response	Customers want a seamless payment process where they get the order when the amount is deducted from their account
Customers find the payment experience is long	The customer waits for payment page to load and respond without any indicator	Customers do not want a long payment process
Customers have a negative impression of the application due to payment-related issues.	Payment related concerns have led to high payment cancellations and negative reviews	Customer want to feel like the application supports their interests and is secure for online transactions

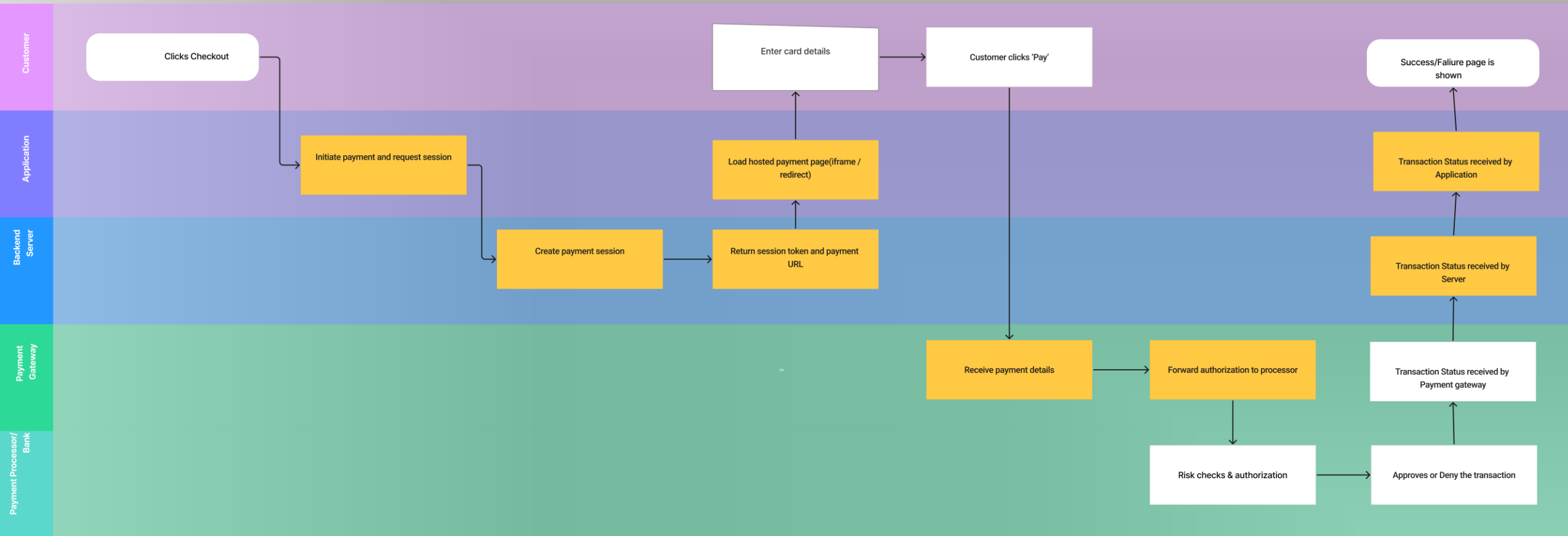
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Key Takeaway: Payment is a complex process with multiple parties involved.

Suppliers	Inputs	Process	Outputs	Customers
Customers & Testers	Customer clicks checkout	Customer clicks checkout	Response from Payment gateway	Customers & Testers
Payment Gateway	Transaction initiated with the Payment gateway	Application calls the server to initiate payment and initialises an iframe		
Payment Processor		Server starts sessions with payment gateway and returns link		
Application		Application opens link in the iframe		
		Customer enters card details in payment and clicks proceed	Customer Requirements • Session time: target response time of 3 minutes • Payment acceptance rate: Target rate is 85% • Communication of payment status is clear	
		The application and the server waits for a response regarding the session		
		Payment gateway uses card details to collect amount from customer card and via the payment processor		
		Payment gateway returns if payment was processed successfully or not.		
		The order is accepted if the payment is successful		

Swimlane Diagram – Current State

Key Takeaway: Payment sessions are handled by the backend server first while the application waits for a response.





Measure Phase

Data Collection Plan

Measures	Data Type	Operational Definition	Stratification Factors	Sampling Notes	Who and How
Payment Session Time	Seconds Continuous	The time taken from the start of the session to the end (either cancelled by the customer or a response is given by payment gateway	Device Type (Android, iOS)	Sampling every transaction for one week	The developer will share logs with the Data Analyst to collect data
Payment Response Time	Seconds Continuous	The time for the payment session after being initiated or the payment response from the payment gateway	Device Type (Android, iOS)	Sampling every transaction for one week	The developer will share logs with the Data Analyst to collect data
Payment Cancellation rate	Percentage	Payment was cancelled by the customer at Payment Gateway page	Device Type (Android, iOS)	Sampling every transaction and aggregating data for the last 2 months	Data Analyst to collect data based on order data
Payment Pending rate	Percentage	Payment response was not received from Payment Gateway page	Device Type (Android, iOS)	Sampling every transaction and aggregating data for the last 2 months	Data Analyst to collect data based on order data
Payment acceptance rate	Percentage	Payment was successfully completed by the Customer.	Device Type (Android, iOS)	Sampling every transaction and aggregating data for the last 2 months	Data Analyst to collect data based on order data
Overall Satisfaction Score	Discrete	Measure of how well the platform meets or exceeds customer expectations across the entire shopping journey	None	Customer feedback survey	The marketing team will collect survey results and share with the data analyst
Net Promotor score	Discrete	Measure of customer loyalty by asking how likely they are to recommend a company on a 0-10 scale.	None	Customer feedback survey	The marketing team will collect survey results and share with the data analyst

Measurement System description

Key Takeaway: Data is reproducible and accurate to system definitions.

Element	Description
Data Source	Application logs & payment gateway logs
Measurement Type	Automated system timestamps & Captured transaction data
Data Owner	Engineering / Platform team
Human Input	None at collection, Aggregation of data
Measurement Frequency	Real-time per transaction
Measurement Risk	Low (system-generated)

Measurement System Validation

Accuracy	Repeatability	Reproducibility	Stability
Timestamps captured by API logs with system boundaries (request sent / response received)	Same transaction ID produces consistent duration when recalculated	Metrics computed independently by: • Application logs • Gateway logs	No structural change to logging during baseline period
Clock synchronization validated with backend server linked to UTC time	Requiring Payment gateway logs collected at PG dashboard supports the collected data	Aggregation of data is done using tools with standard formulas	Metric definitions remained constant

Baseline

- 1. “For e-commerce platforms, the average authorization rate typically ranges between 85% and 95%”- [Gr4vy](#)
- 2. “Globally, card payments fail 10-15% of the time.” - [Riveo](#)
- 3. A good checkout abandonment rate is lower than 40%. It is argued that the best optimized abandonment rate is approximately 20%,” - [Bolt](#)

Key Takeaway: Customers are not satisfied with the application experience.

3.1 / 5
Overall App Satisfaction

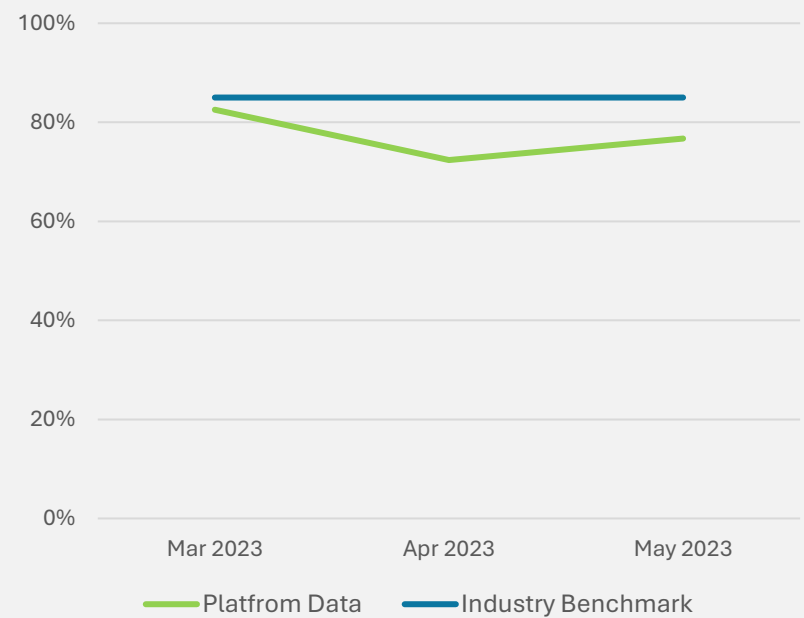
4 / 10
Net Promotor Score (NPS)

77% (vs) 80%
Payment Acceptance
Platform (VS) Benchmark (1)

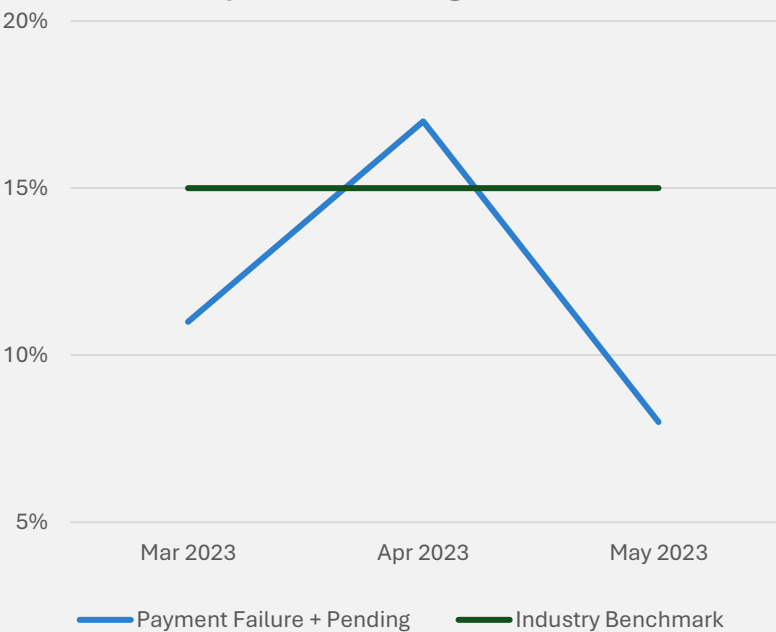
12% (vs) 10%
Payment failure + Pending
Platform (VS) Benchmark (2)

10% (vs) 20%
Payment Cancelled (Abandoned Payments)
Platform (VS) Benchmark (3)

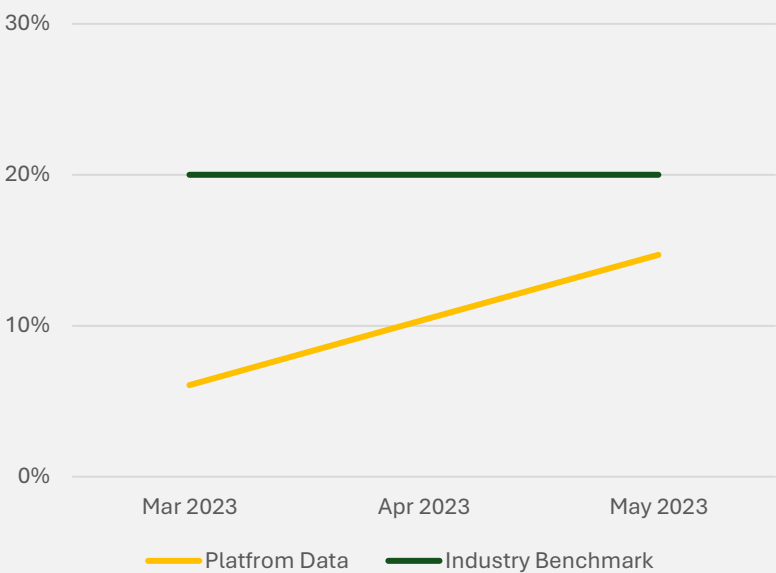
Payment Acceptance



Payment Pending + Failure



Payment Cancelled
(Checkout Abandonment)



Period: March 2023 – May 2023

Baseline

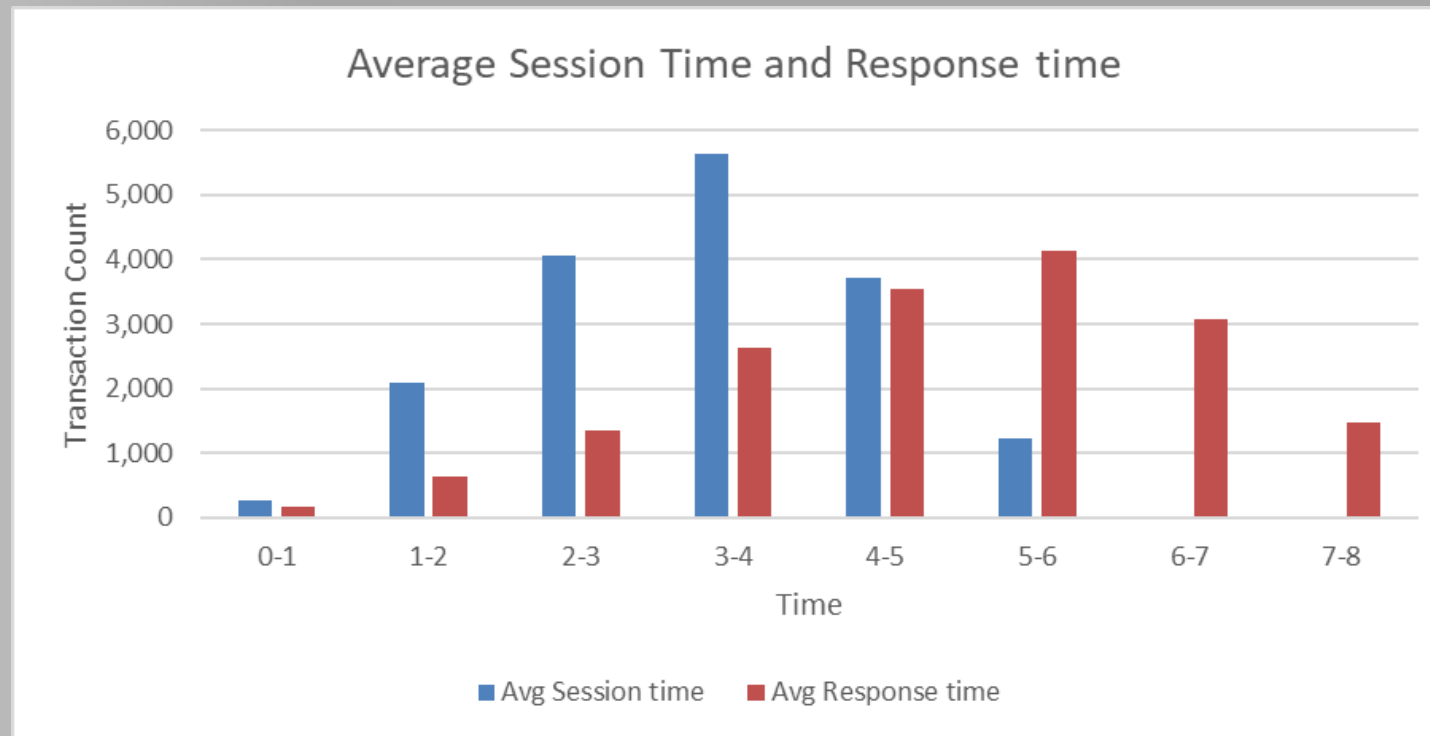
Key Takeaway: Customers are waiting an additional minute for the payment response, leading to them cancelling payments.

4.3 Mins

Average Session time

5.30 Mins

Average Payment response time

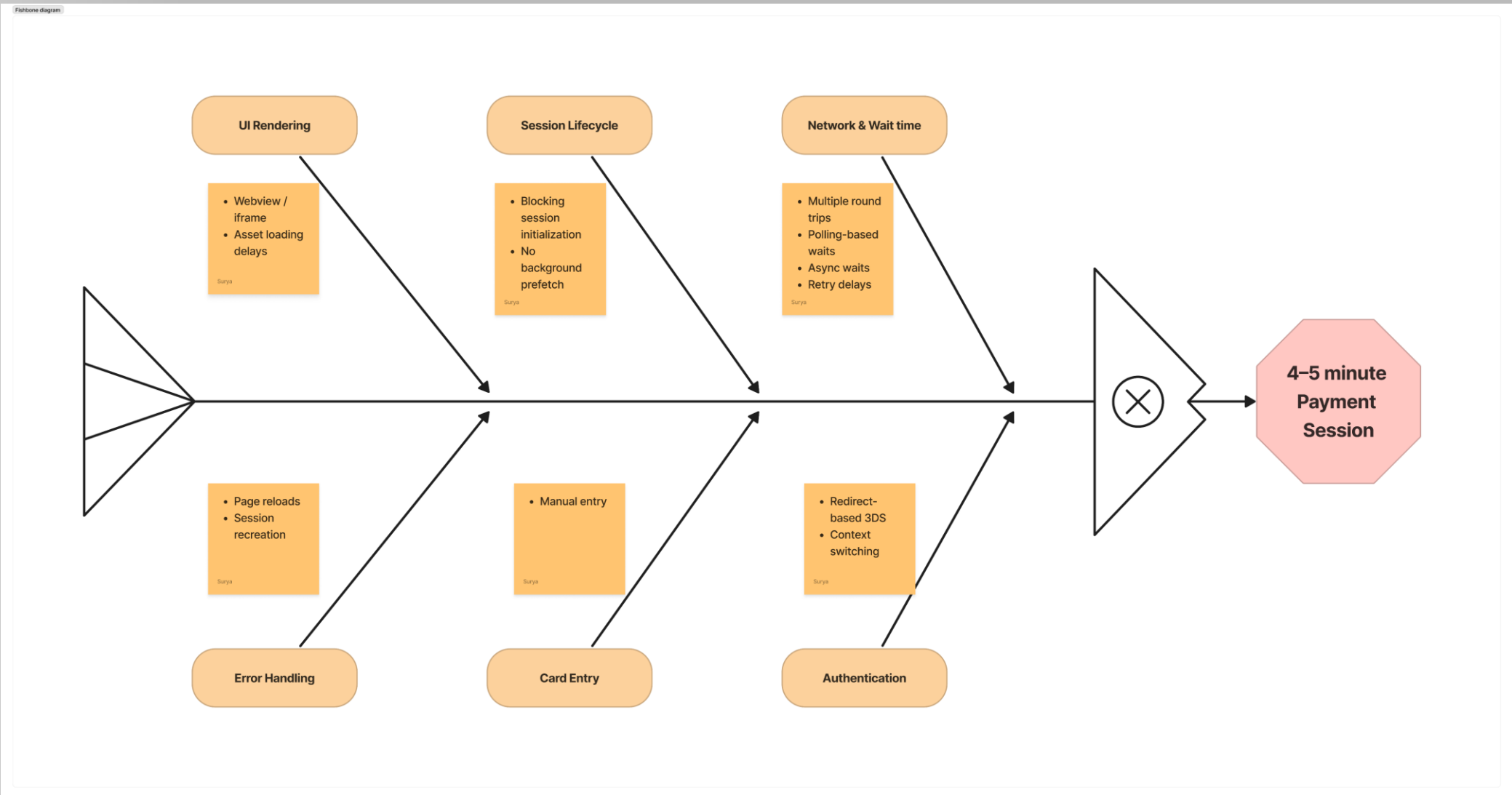




Analyze Phase

Fishbone Diagram

Key Takeaway: Multiple causes can lead to long payment sessions.



5 Whys

Key Takeaway: All the load for payment is handled by the Application server.

	Why 1	Why 2	Why 3	Why 4	Why 5
Why	Why are customers frustrated with our payment process	Why do they feel this sentiment?	What is causing the average payment response time	Why is load of payment being handled by server?	Why are sessions being initiated by the server?
Because	Customers feel the payment process takes too long and is not reliable.	The average response from the payment gateway takes time, and the customer has to wait till this response to proceed.	Network traffic can cause packet loss, a routing mismatch or server response and processing time . Missed responses are not captured by re-request.	The server is considered the single source of truth and is used to handle all backend transactions such as payment.	The complete payment lifecycle is being managed by the server as a backend service.

Value Add & Cycle Time Analysis

Key Takeaway: Time is lost in initiating the session, setting up UI and backend processing.

#	Activity	Description	Avg Time (sec)	Classification	Why
1	Checkout initiated	User clicks “Checkout”	5	VA	Directly progresses toward payment
2	Session initialization	Create payment session, token, iframe	35	NVA	Required but no customer-perceived value
3	UI load & render	Payment UI / iframe loads	25	NVA	Technical necessity
4	Customer enters card details	PAN, expiry, CVV, name	95	VA	Customer actively completes payment
5	Client-side validation	Luhn check, field validation	10	NVA	Prevents errors but adds no value
6	3DS / OTP authentication	Bank verification step	45	VA	Security step required to complete payment
7	Authorization request	Sent to gateway / processor	15	NVA	Backend processing
8	Processor decisioning	Risk checks, auth decision	20	NVA	Required for transaction
9	Waiting for response	Network / async wait	10	NVA R-time	Pure wait
10	UI confirmation rendering	Success / failure screen	10	NVA	Informational only

Category	Total Time (sec)	% of Session
Value-Added (VA)	145 sec	~54%
NVA (Required)	105 sec	~39%
NVA R-time (Waiting/Rework)	10 sec	~4%
Total	260 sec (~4.3 min)	~100%

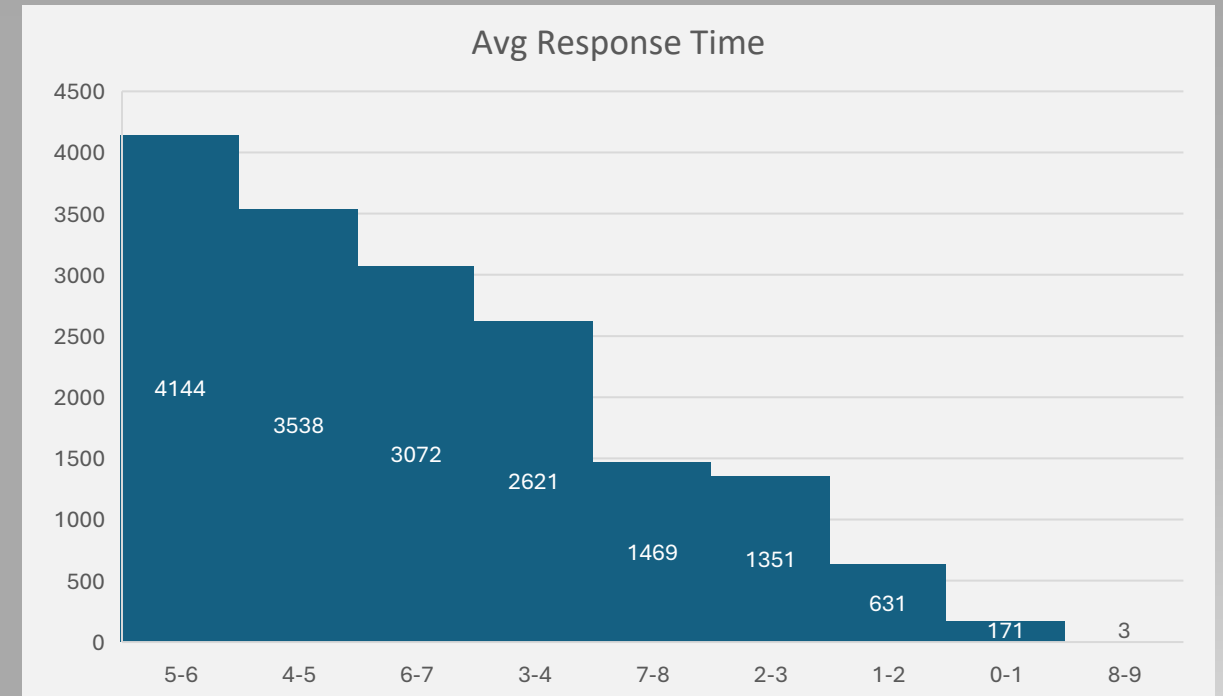
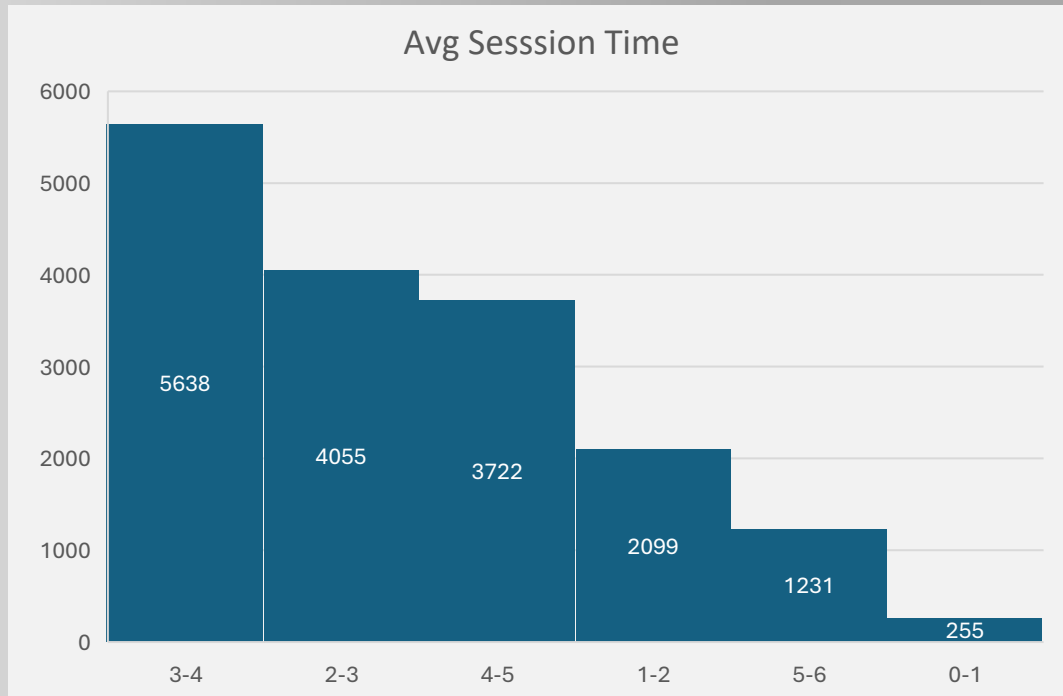
Root Cause Analysis

Key Takeaway: Webhosted Checkout method adds constraints compared to SDK native payments.

Hypothesis #	Possible Root Cause (x)	Root Cause Hypothesis	Result
1	UI rendering time	Payment UI is loaded via iframe, requiring HTML, JS, and CSS assets to be fetched at runtime, significantly delaying checkout start.	TRUE
2	Session initialization timing	Payment session is created only after checkout is initiated, placing session creation on the critical path and increasing total session duration.	TRUE
3	Network round trips	Web-hosted checkout requires multiple client-server-gateway hops and polling mechanisms, increasing wait time and latency variance.	TRUE
4	Card entry method	Web-based card forms do not leverage OS-level autofill or native controls, increasing customer input time and error rates.	TRUE
5	3DS authentication flow	Redirect-based or iframe 3DS flows introduce context switching and additional rendering delays compared to in-app authentication.	TRUE
6	Error handling & retries	Failures trigger page reloads and session recreation in web-hosted checkout, inflating session time due to rework and idle waits.	TRUE
7	Backend processor latency	Payment processor authorization and risk checks are slower than acceptable thresholds and are the primary contributor to long session times.	FALSE

Hypothesis confirmations

Key Takeaway: The average payment session ends before the payment response is received from the payment gateway.

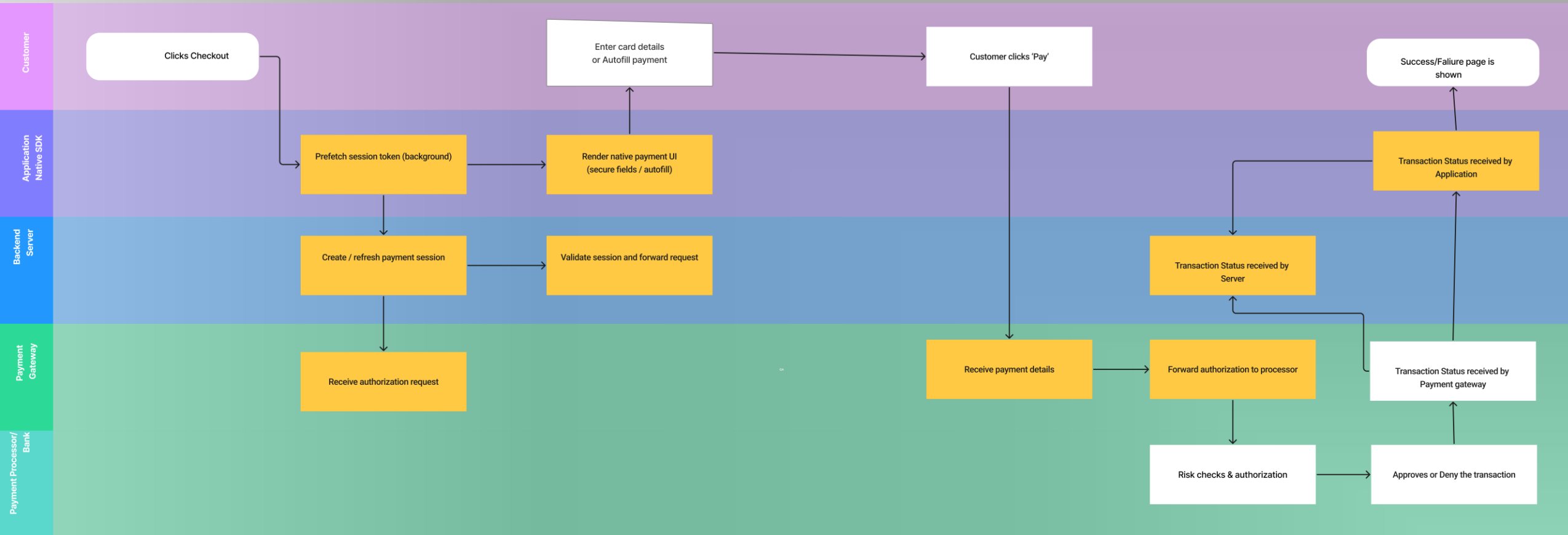




Improve
Phase

Swimlane Diagram – New State

Key Takeaway: Payment sessions are handled by the Native SDK while keeping the server informed.



Hypothesis confirmations

Key Takeaway: Moving to native SDK integration for payments with optimizations for network.

#	Solution Option	Impact to Customer (1–5)	Potential to Meet Goal (1–5)	Effort by Team (1–5)	Time to Implement (1–5)	Total Weighted Score
Weight		30%	30%	20%	20%	$(0.30 \times \text{Impact}) + (0.30 \times \text{Goal Potential}) - (0.20 \times \text{Effort}) - (0.20 \times \text{Time})$
1	Native SDK checkout	5	5	4	5	1.2
2	Session prefetch & reuse	4	4	2	2	1.6
3	Native autofill / wallets	4	4	2	1	1.8
4	Reduce network round trips and webhooks for payment response	4	4	3	3	1.2
5	In-app 3DS (biometric)	3	3	3	3	0.6
6	Processor replacement	1	1	5	5	-1.8

Value Add & Cycle Time Analysis

Key Takeaway: Initiating the session, setting up UI and backend processing via native SDK.

#	Activity	Description	Avg Time (sec)	Classification	Why
1	Checkout initiated	User taps “Pay”	3	VA	Direct customer intent
2	Session/token prefetch	Session created in background	15	NVA	Required for security
3	Native payment UI render	SDK UI loads instantly	5	NVA	Technical setup
4	Card entry (native fields)	PAN, expiry, CVV, name	55	VA	Customer completes payment
5	Native validation	Inline field validation	5	NVA	Prevents errors
6	Wallet / autofill (optional)	Apple Pay / saved card	10	VA	Accelerates checkout
7	3DS in-app challenge	Biometric / OTP	25	VA	Security + completion
8	Authorization request	Sent to gateway	10	NVA	Backend processing
9	Processor decisioning	Risk & auth checks	12	NVA	Compliance
10	Response wait	Network round-trip	5	NVA R-time	Pure wait
11	UI confirmation	Success / failure screen	5	NVA	Informational

Category	Total Time (sec)	% of Session
Value-Added (VA)	93 sec	~62%
NVA (Required)	52 sec	~35%
NVA R-time (Waiting)	5 sec	~3%
Total	150 sec (~2.5 min)	~100%

Testing Data to Validate Impact

Key Takeaway: The average payment session and response time align, with differences for 6% of transactions. These transactions are captured via webhook, and the communication is sent to the customer.

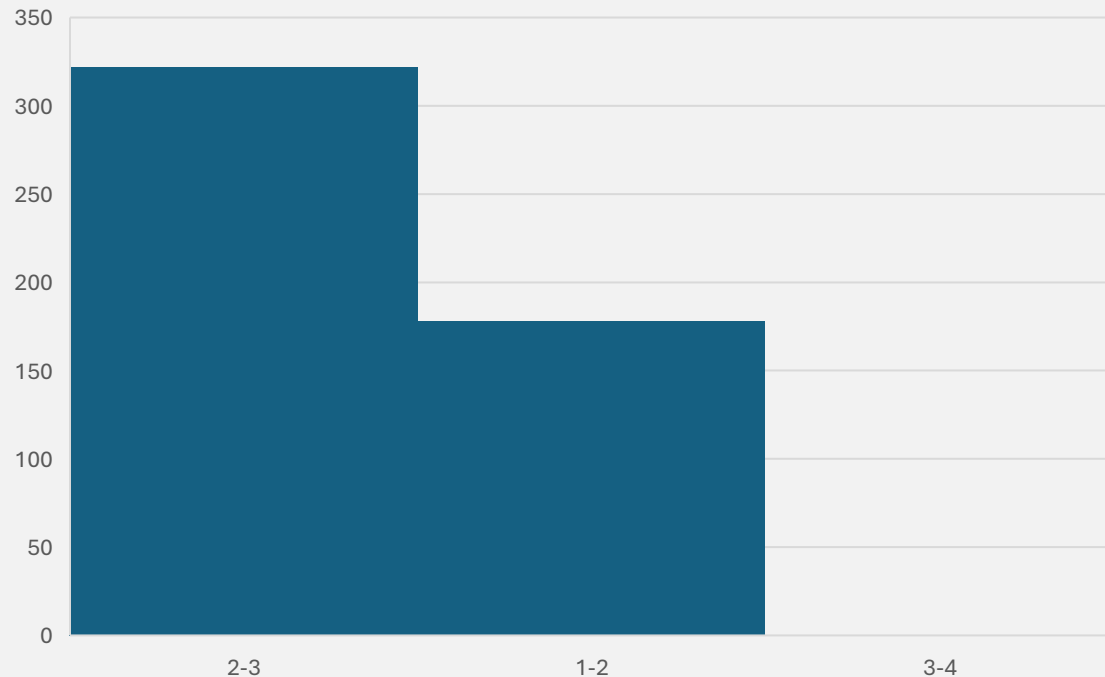
2.5 Mins

Average Session time

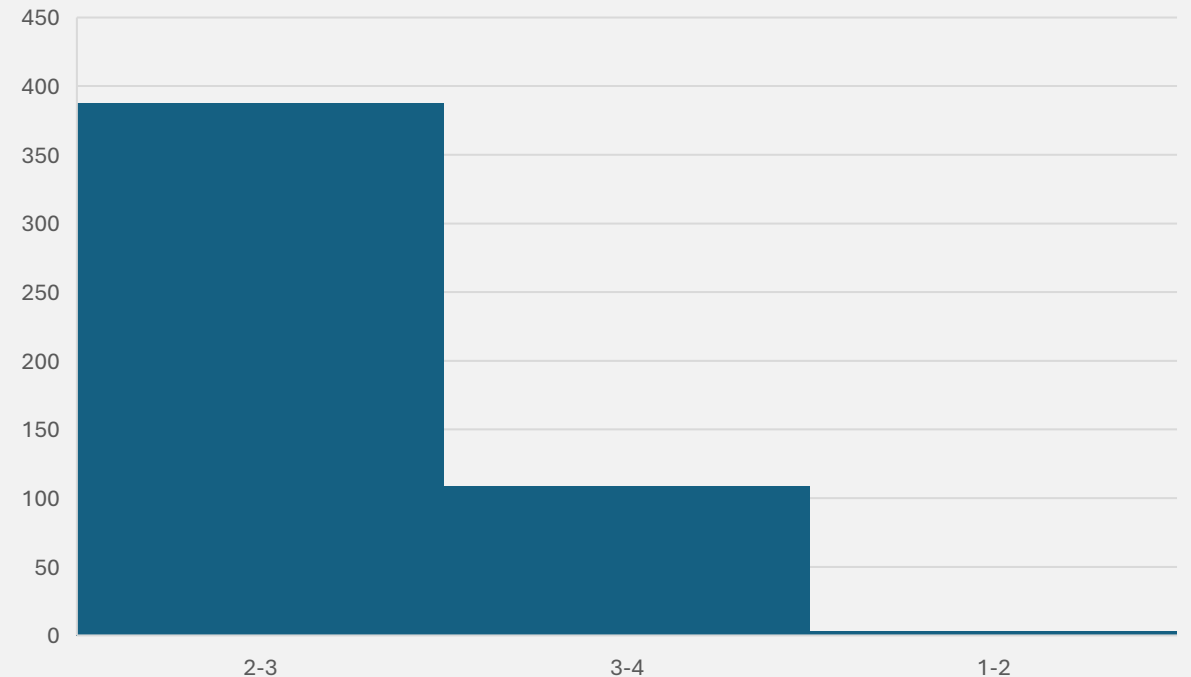
2.78 Mins

Average Payment response time

Avg Session time



Avg Response time



Improvement after Deployment

Key Takeaway: Customers are experiencing a better payment journey.

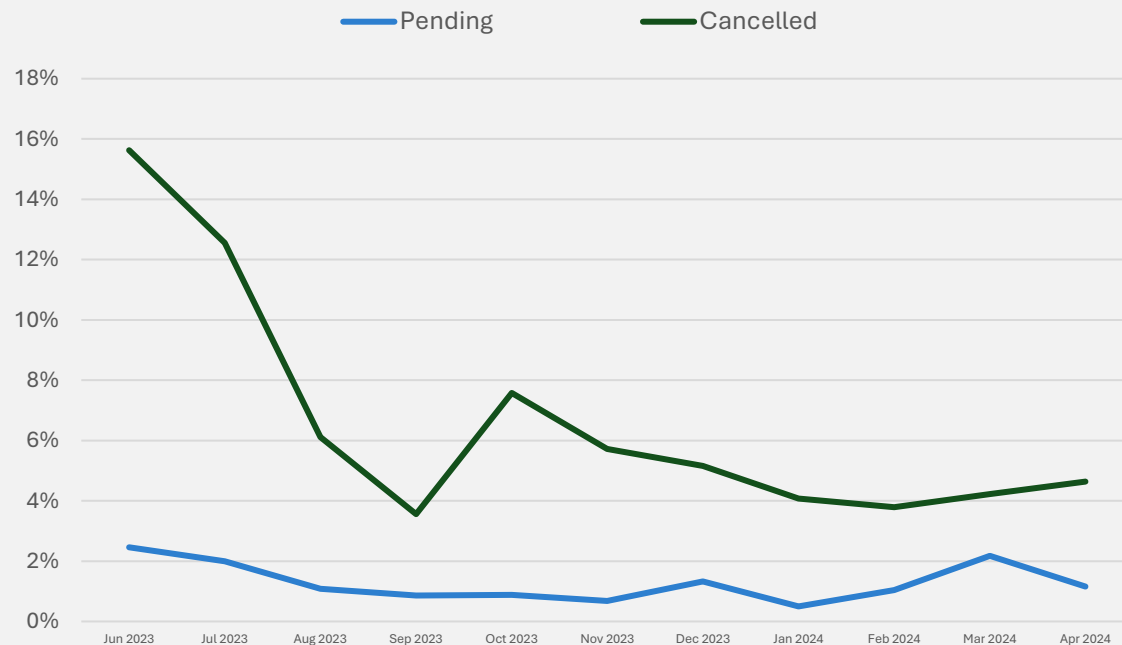
2.52 Mins

Average Session time

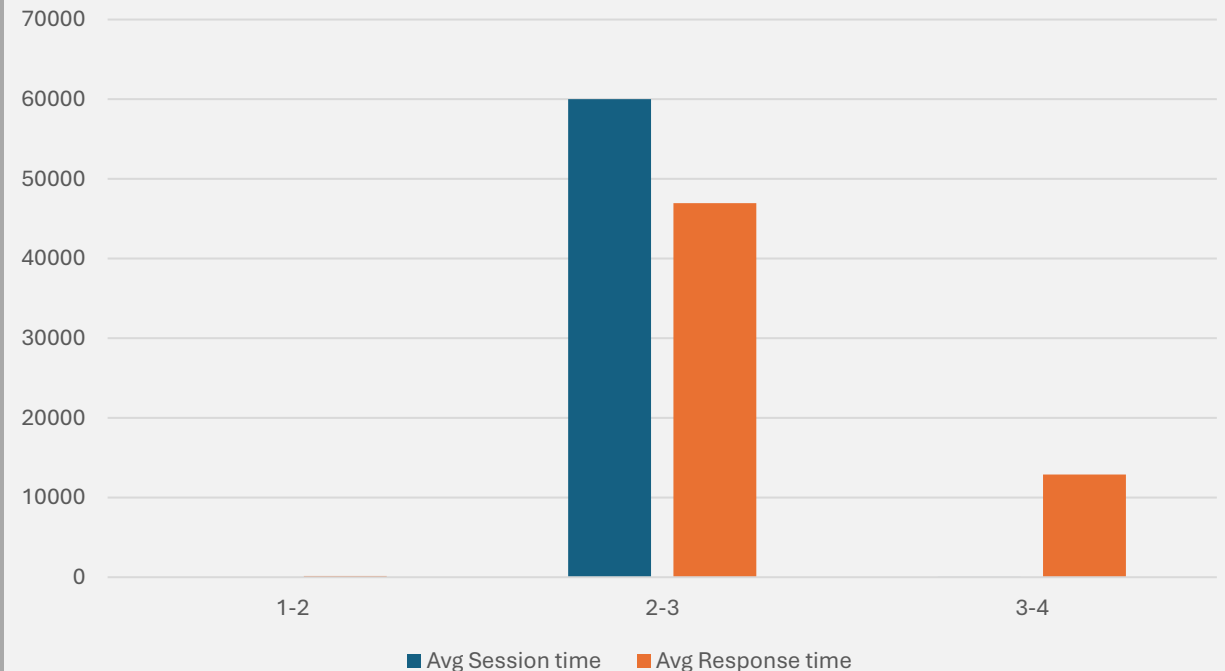
2.79 Mins

Average Payment response time

Payment Failure Vs Pending



Average Session Time and Response time



Improvement after Deployment

1. "For e-commerce platforms, the average authorization rate typically ranges between 85% and 95%" - [Gr4vy](#)
2. "Globally, card payments fail 10-15% of the time." - [Riveo](#)
3. A good checkout abandonment rate is lower than 40%. It is argued that the best optimized abandonment rate is approximately 20%," - [Bolt](#)

Key Takeaway: App Ratings have improved after changes.

4.2 / 5

Overall App Satisfaction

7 / 10

Net Promotor Score (NPS)

92% (vs) 85%

Payment Acceptance
Platform (VS) Benchmark (1)

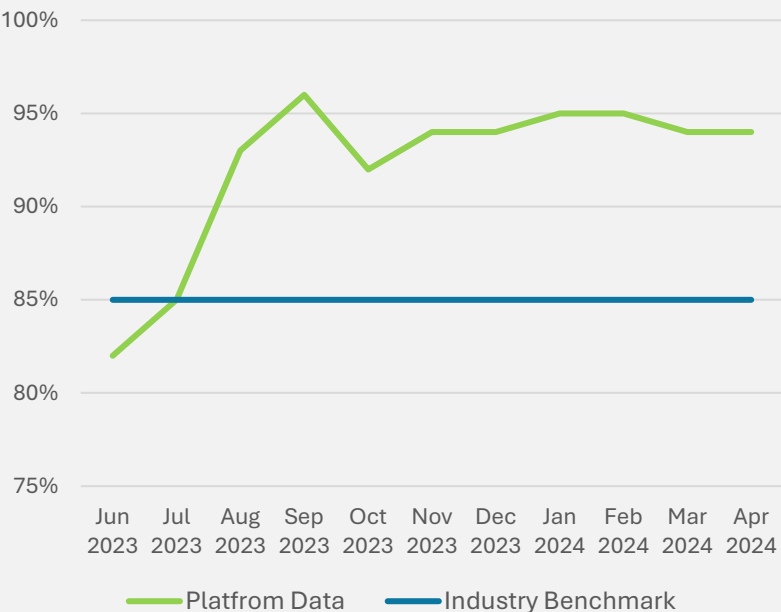
1% (vs) 10%

Payment failure + Pending
Platform (VS) Benchmark (2)

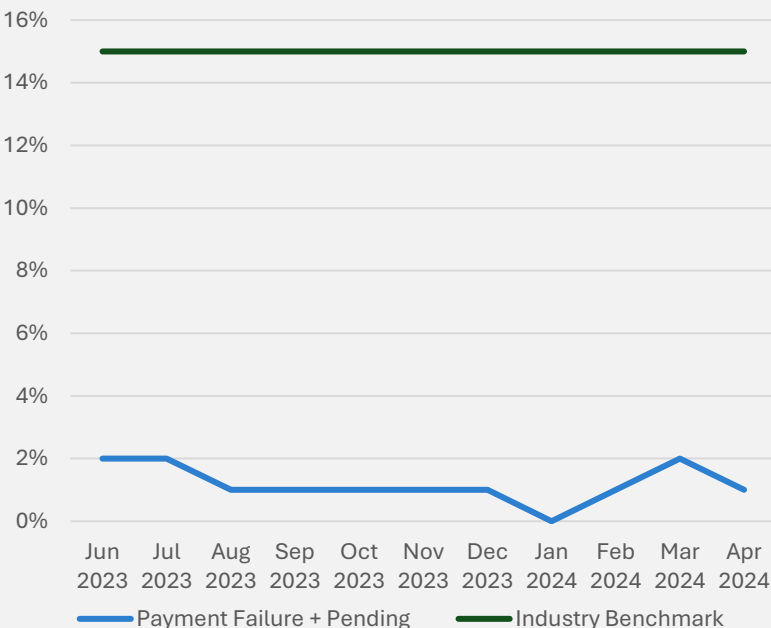
7% (vs) 20%

Payment Cancelled (Abandoned Payments)
Platform (VS) Benchmark (3)

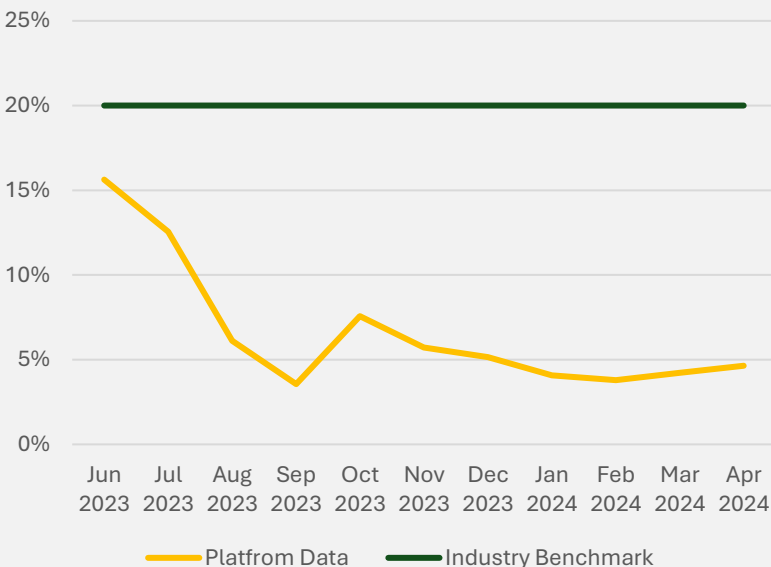
Payment Acceptance



Payment Pending + Failure



Payment Cancelled
(Checkout Abandonment)



Period: March 2023 – May 2023



Control
Phase

Project Closure

Lessons Learned

- Payment is a critical factor for the adoption of ecommerce experience
- Over-reliance on the server to handle logic can affect the performance of the platform.

Customer Impact

- Customer Overall satisfaction increased to 4.2
- Customers are more likely to recommend application with NPS score of 7
- Payment KPIs are above expected benchmarks with Payment completion at 92%
- Customers cancelling transactions at payment dropped by 3%

Calculations of Gains

Hard Savings/Profit	Soft Savings
Increase in sales by 7% per month	Increase customer satisfaction leading to more frequent purchases (Average frequency of orders per month per customer is at 1.4)
	Improved reputation of the application and brand influence
	Decreased customer complaints regarding payments by 67%

Process Handover

Hard Savings/Profit	Soft Savings
Has the new SDK implementation been documented for future reference	Yes
Has the KPIs setout for the project being met	Yes
Has the dashboard been set up with alerts for tracking payment journey anomalies	Yes

Value Add & Cycle Time Analysis

Key Takeaway: Ensure the payment process is healthy by reviewing the aggregate session KPIs.

Name of the Measure	Input, Process or Output?	What is the Target?	Method of Data Capture	Checking Frequency	Person Responsible	Measure	Upper / Lower Trigger Point	Who Will Respond?	Reaction Plan
Checkout Session Time (SDK)	Output	≤ 3.5 minutes avg	Timestamp difference between SDK session start and completion	Daily	Data Analyst	Checkout Session Time	> 4.5 min avg	Release Lead	Review SDK logs, validate session prefetch, check UI rendering delays
Payment Response Time	Output	≤ 3.5 minutes avg	Gateway callback timestamps	Daily	Data Analyst	Payment Response Time	> 4.5 min avg	Release Lead	Validate gateway latency, network routing, retry behavior
p95 Checkout Duration	Output	≤ 4 minutes	Aggregated latency metrics	Weekly	Data Analyst	p95 Checkout Duration	> 5 min	Engineering Lead	Deep-dive on tail latency, retries, 3DS flow, and network round trips
SDK Session Initialization Time	Process	≤ 20 seconds	SDK instrumentation logs	Weekly	Data Analyst	SDK Init Time	> 30 sec	App Developer	Inspect token prefetch, cold starts, SDK caching
Network Round Trips per Transaction	Process	≤ 3 calls	API gateway & client telemetry	Weekly	Network Engineer	Network Round Trips	> 4 calls	Network Engineer	Identify redundant API calls, batching opportunities
Checkout Failure Rate	Output	≤ 1%	Payment failure events	Daily	Data Analyst	Checkout Failure Rate	> 2%	Backend Developer	Trigger incident review, enable fallback path, notify gateway

Risk Management

Key Takeaway: Ensure proper error handling and alerts to keep customer experience consistent.

Process Step	Failure Mode	Potential Effect	Potential Cause	Severity	Occurrence	Detection	RPN	Current Controls	Recommended Action
SDK session initialization	Session not prefetched	Increased checkout time	Prefetch logic not triggered	7	4	4	112	SDK logs, metrics	Enforce prefetch at app start; add guardrails
SDK session initialization	Token expiry mid-session	Checkout failure	Token TTL too short	8	3	3	72	Gateway validation	Increase TTL; auto-refresh tokens
Network calls	Excess round trips	Latency inflation	Redundant API calls	6	5	4	120	API telemetry	Batch calls; eliminate polling
Card entry	Autofill not triggered	Longer input time	OS permissions / configurations	5	4	3	60	UX testing	Add fallback UX; validate device support
3DS authentication	3DS challenge fails	Payment drop-off	SDK–issuer mismatch	9	3	4	108	Gateway monitoring	SDK updates; issuer compatibility testing
SDK retries	Retry loop	Duplicate charges / delays	Missing idempotency	10	2	5	100	Idempotency keys	Enforce strict idempotency
Error handling	Silent SDK error	Customer abandonment	Poor error surfacing	8	3	6	144	Logs only	User-facing error + analytics
Deployment	Partial rollout	Mixed experiences	Feature flag misconfiguration	7	2	3	42	Release checklist	Automated rollout validation
Monitoring	Anomaly spike unnoticed	Delayed response	Alert thresholds too high	9	2	6	108	Dashboards	Lower alert thresholds; PagerDuty



Appendix

Key Words

Term	Definition
Application	An e-commerce application developed for brands with specialised online ordering.
Server	Backend channel that handles platform logic and data collection.
Customer	End user using the application and placing an order
Payment acceptance rate	Payment was successfully completed by the Customer
Payment pending rate	Payment response was not received from Payment Gateway page
Payment cancelled/Abandoned rate:	Payment was cancelled by the customer at Payment Gateway page
Middleware	
Payment Gateway	Third-party provider middleware that handles payment between the application and the payment processor.
Payment processor	Third-party provider that handles the transaction between the customer and the organization's accounts.
Payment Session	A temporary, stateful object created to track a payment attempt from initiation to completion.
Session Prefetch	The background creation of a payment session before the customer reaches checkout.
3DS / 3D Secure	An additional authentication step required by issuers to validate cardholder identity.
Webhosted Checkout	A payment flow that loads an external payment page via iframe or redirect within an app or website.
Native SDK checkout integration	A payment integration using a mobile platform's native software development kit instead of a web-based checkout page.
Network Round trip	A full request-and-response communication between client and server.
Redirect-Based Authentication	An authentication method that navigates the user away from the app to complete verification.
iframe	An embedded web page within another page, often used to host third-party content.